

CLASS :
EXAM NO :

TIME : 3 Hrs
MARKS : 80

SECTION A (1 mark each) (14 × 1 = 14)

- 1) Ctenophores are commonly known as:
 - a) sea walnuts
 - b) comb jellies
 - c) both a and b
 - d) sea anemones
- 2) Which of the following is a characteristic feature of phylum Ctenophora?
 - a) Metamerism
 - b) Radial symmetry
 - c) Bioluminescence
 - d) Pseudocoelom
- 3) The body of ctenophores bears how many external rows of ciliated comb plates?
 - a) Four
 - b) Six
 - c) Eight
 - d) Ten
- 4) Digestion in ctenophores is:
 - a) Only extracellular
 - b) Only intracellular
 - c) Both extracellular and intracellular
 - d) Neither extracellular nor intracellular
- 5) Which of the following is NOT a characteristic of phylum Ctenophora?
 - a) Exclusively marine organisms
 - b) Radial symmetry
 - c) Tissue level of organization
 - d) Triploblastic body plan
- 6) In ctenophores, the property of a living organism to emit light is called:
 - a) Bioluminescence
 - b) Fluorescence
 - c) Phosphorescence
 - d) Chemiluminescence
- 7) Ctenophores exhibit which type of body symmetry?
 - a) Bilateral symmetry
 - b) Radial symmetry
 - c) Asymmetry
 - d) Pentamerous symmetry
- 8) Which of the following helps in locomotion in ctenophores?

- a) Tentacles
 - b) Cilia
 - c) Ciliated comb plates
 - d) Parapodia
- 9) The level of organization in ctenophores is:
- a) Cellular level
 - b) Tissue level
 - c) Organ level
 - d) Organ-system level
- 10) In dicotyledonous plants, the vascular bundles are:
- a) Scattered
 - b) Arranged in a ring
 - c) Located in the center
 - d) Absent
- 11) The stem of monocotyledonous plants is characterized by:
- a) Vascular bundles arranged in a ring
 - b) Presence of pith
 - c) Scattered vascular bundles
 - d) Presence of secondary growth
- 12) In dorsiventral leaves, the palisade parenchyma is present towards:
- a) Lower epidermis
 - b) Upper epidermis
 - c) Both upper and lower epidermis
 - d) Middle of the leaf
- 13) The male reproductive organ in frogs is:
- a) A single testis
 - b) A pair of testes
 - c) Multiple testes
 - d) Gonads without definite shape
- 14) How many ova can a mature female frog lay at a time?
- a) 250-300
 - b) 500-1000
 - c) 1000-2000
 - d) 2500-3000

SECTION B (2 marks each) (10 × 2 = 20)

- 15) What are the distinguishing features of ctenophores from other marine organisms?
Ctenophores are exclusively marine, radially symmetrical, diploblastic organisms with tissue level of organisation. Their body bears eight external rows of ciliated comb plates which help in locomotion, and they exhibit bioluminescence.
- 16) How does digestion occur in ctenophores?
Digestion in ctenophores is both extracellular and intracellular. The food is broken down outside the cells by digestive enzymes and the partially digested food is then taken up by cells for further

breakdown.

17) What is the significance of water vascular system in echinoderms?

The water vascular system is the most distinctive feature of echinoderms which helps in locomotion, capture and transport of food, and respiration. It's a hydraulic system unique to this phylum that enables their movement.

18) How do phyllotaxy patterns differ in plants?

In alternate phyllotaxy, a single leaf arises at each node in alternate manner as in china rose. In opposite phyllotaxy, a pair of leaves arise at each node and lie opposite to each other as in Calotropis. In whorled phyllotaxy, more than two leaves arise at a node and form a whorl, as in Alstonia.

19) What are the characteristics of a monocotyledonous seed?

Monocotyledonous seeds are generally endospermic with a membranous seed coat fused with the fruit wall. The embryo is small with one large shield-shaped cotyledon called scutellum, and a short axis with plumule and radicle enclosed in protective sheaths called coleoptile and coleorhiza respectively.

20) How are vascular bundles arranged differently in monocot and dicot stems?

In dicot stems, vascular bundles are arranged in a ring pattern with conjoint, open bundles having cambium between xylem and phloem. In monocot stems, vascular bundles are scattered throughout the ground tissue, are conjoint and closed (without cambium), and each is surrounded by a sclerenchymatous bundle sheath.

21) What is the role of bulliform cells in grass leaves?

Bulliform cells are large, empty, colorless cells found in the adaxial epidermis along the veins in grass leaves. When these cells are turgid with water, the leaf surface is exposed, but when they become flaccid due to water stress, they cause the leaves to curl inwards to minimize water loss.

22) How is cutaneous respiration carried out in frogs?

In cutaneous respiration, the skin of the frog acts as an aquatic respiratory organ. The skin is highly vascularized and kept moist by mucus secretions, allowing dissolved oxygen in water to be exchanged through the skin by diffusion when the frog is submerged.

23) What is the function of the renal portal system in frogs?

The renal portal system in frogs is a special venous connection between the kidney and lower parts of the body. It carries blood from the posterior parts of the body to the kidneys before returning to the heart, allowing efficient filtration of waste products.

24) How do male and female frogs differ externally?

Male frogs can be distinguished by the presence of sound-producing vocal sacs and a copulatory pad on the first digit of the forelimbs, which are used during mating. These features are absent in female frogs, demonstrating sexual dimorphism in the species.

SECTION C (3 marks each) (7 × 3 = 21)

25) Describe the characteristic features of phylum Ctenophora. How do they differ from Cnidarians?

Ctenophores, commonly known as sea walnuts or comb jellies, are exclusively marine, radially symmetrical, diploblastic organisms with tissue level of organization. The body bears eight

external rows of ciliated comb plates, which help in locomotion. Digestion is both extracellular and intracellular. Bioluminescence is well-marked in ctenophores. Unlike cnidarians, ctenophores lack cnidoblasts or stinging cells, and they have complete digestive systems with separate mouth and anal openings.

26) Explain the water transport system in sponges. How does this help them in their survival?

In sponges, water enters through minute pores (ostia) in the body wall into a central cavity called spongocoel, from where it goes out through the osculum. This pathway of water transport is facilitated by the beating of flagella in specialized cells called choanocytes that line the spongocoel and canals. This water transport system is helpful in food gathering as food particles are filtered from the incoming water, respiratory exchange of gases occurs through the flowing water, and waste materials are removed through the outgoing water, ensuring their survival.

27) What is radial symmetry? Name three animal phyla that exhibit this type of symmetry and explain its significance.

Radial symmetry is a body plan where any plane passing through the central axis of the body divides the organism into two identical halves. Phyla exhibiting radial symmetry include Coelenterata (Cnidaria), Ctenophora, and adult Echinodermata. Radial symmetry is significant as it allows organisms to respond to stimuli from all directions equally, which is advantageous for sessile animals like sea anemones or free-floating organisms like jellyfish that need to capture food or respond to threats from any direction in their aquatic environment.

28) Describe the structure and function of stomata. Why are they more numerous on the lower surface of dicot leaves?

Stomata are specialized structures in the epidermis of leaves consisting of two bean-shaped or dumb-bell shaped guard cells which enclose a stomatal pore. The guard cells contain chloroplasts and regulate the opening and closing of the stomatal pore. Stomata regulate transpiration and gaseous exchange between the plant and atmosphere. They are more numerous on the lower surface of dicot leaves to reduce water loss, as the lower surface is less exposed to direct sunlight and heat, while still allowing sufficient gas exchange for photosynthesis.

29) Compare and contrast the vascular bundles found in dicot stems and monocot stems.

In dicot stems, vascular bundles are arranged in a ring pattern and are conjoint, open, and with endarch protoxylem. They have cambium between xylem and phloem that enables secondary growth. In monocot stems, vascular bundles are scattered throughout the ground tissue and are conjoint, closed (without cambium), and surrounded by a sclerenchymatous bundle sheath. Peripheral vascular bundles in monocots are generally smaller than the centrally located ones, and phloem parenchyma is absent. Additionally, monocot bundles often contain water-filled cavities.

30) Describe the digestive system of frog. How is it adapted to its carnivorous habit?

The digestive system of frog consists of a short alimentary canal and digestive glands. It includes mouth, buccal cavity, pharynx, esophagus, stomach, duodenum, intestine, rectum, and cloaca. Digestive glands include liver, gallbladder, and pancreas. The system is adapted to a carnivorous habit by having a short intestine (as animal food is easier to digest than plant material), a muscular bilobed tongue for capturing prey, well-developed gastric glands that secrete HCl and digestive enzymes, and numerous finger-like villi in the intestine for efficient

absorption of nutrients.

31) Explain the respiratory system of frog. How does it adapt to both aquatic and terrestrial life? Frogs have a dual respiratory system that allows them to respire both in water and on land. In water, the highly vascularized skin acts as an aquatic respiratory organ (cutaneous respiration) where dissolved oxygen diffuses through the moist skin. On land, they use pulmonary respiration through a pair of pink, elongated, sac-like lungs in the thorax. Air enters through the nostrils into the buccal cavity and then to the lungs. The buccal cavity also assists in respiration. During aestivation and hibernation, gaseous exchange takes place primarily through the skin, demonstrating their adaptation to varied environments.

SECTION D (4 marks each) (2 × 4 = 8)

32) Identify the phylum to which sea walnuts belong and explain two distinctive features that differentiate them from sea anemones. (2 marks)

33) Though starfish appears to have radial symmetry as adults, they are classified under deuterostomes rather than radiata. Justify this classification based on their developmental and anatomical features. (2 marks)

Question 2 (4 marks)

Case Scenario: Plant Adaptation Study

A botanist examined cross-sections of stems from plants collected from different habitats - a desert plant and an aquatic plant. The images below show their vascular tissue arrangements.

Observation data:

- Plant A showed scattered vascular bundles with no distinct arrangement in rings
- Plant B showed vascular bundles arranged in a ring with a distinct cambium layer
- Plant A had a large amount of ground tissue with air spaces
- Plant B showed secondary thickening and less ground tissue

Questions:

SECTION E (5 marks each) (3 × 5 = 15)

Long Answer Questions (5 marks each)

1) Describe the distinctive features of phylum Ctenophora. How does it differ from other diploblastic animals?

OR

Explain the structure and characteristics of Ctenophores. Why are they placed in a separate phylum despite being diploblastic like Cnidaria?

2) Explain the anatomy of a dicotyledonous stem with the help of a labeled diagram. Describe how it differs structurally from a monocotyledonous stem.

OR

Describe the internal structure of a dorsiventral leaf with a labeled diagram. How is it adapted for efficient photosynthesis?

3) Describe the male reproductive system of frog with a labeled diagram. Explain how it is adapted for external fertilization.

OR

Explain the morphological features and respiratory adaptations of frogs that allow them to live both on land and in water.

ANSWER KEY (SECTION A)

- Q1. (c)
- Q2. (c)
- Q3. (c)
- Q4. (c)
- Q5. (d)
- Q6. (a)
- Q7. (b)
- Q8. (c)
- Q9. (b)
- Q10. (b)
- Q11. (c)
- Q12. (b)
- Q13. (b)
- Q14. (d)